Artificial Intelligence

AIML320/420

Tutorial in week 3

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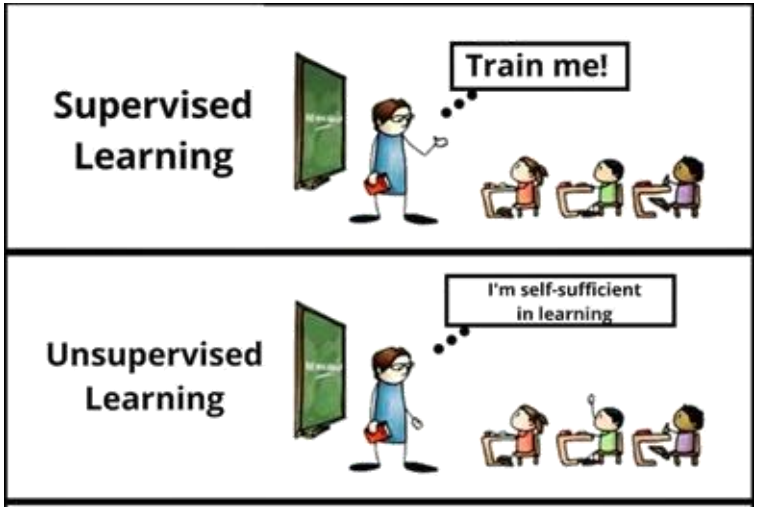
AIML320/420 Week 3

1. Announcements
   - Assignment 1 (**15%**), due on 11:59pm, 26th March (Thur), week 5
   - Helpdesk from week 2: **Tuesday and Thursday 2-3pm at CO242B**
2. First 3 weeks:
   - Supervised learning and unsupervised learning
   - Regression, classification and clustering
   - Dataset split: training, test and validation set
   - Generalisation and overfitting
   - KNN, decision tree, ensemble learning
   - Data preparation
3. Sample codes
4. Assignment 1

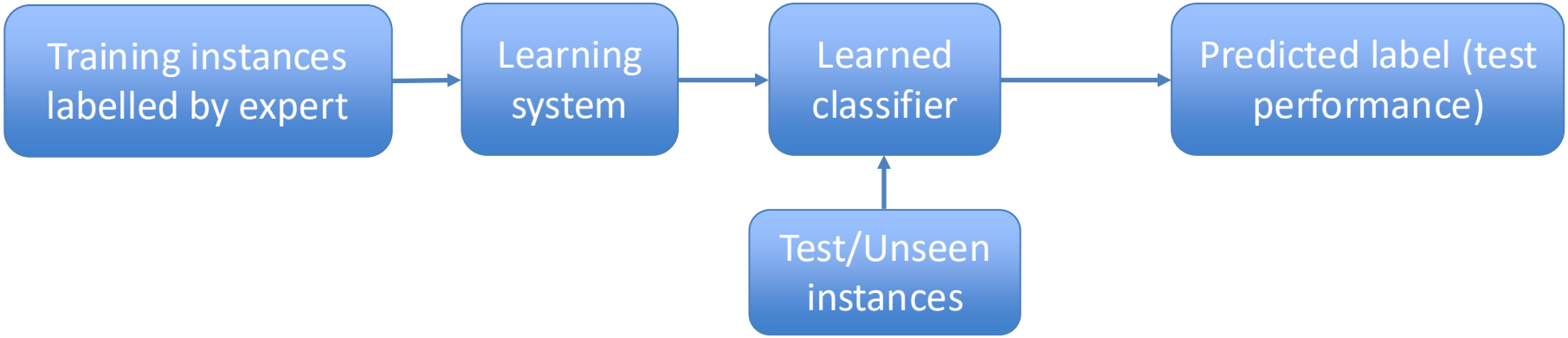
(un)Supervised Learning

- Depends on the data are **labelled (supervised) or not (unsupervised)**
- Difference between supervised and unsupervised learning?
  - Input data is labelled VS input data is unlabelled
  - Classification and regression VS clustering
  - Use training dataset VS use just input dataset
  - Used for prediction VS used for analysis



A Typical Supervised Learning System

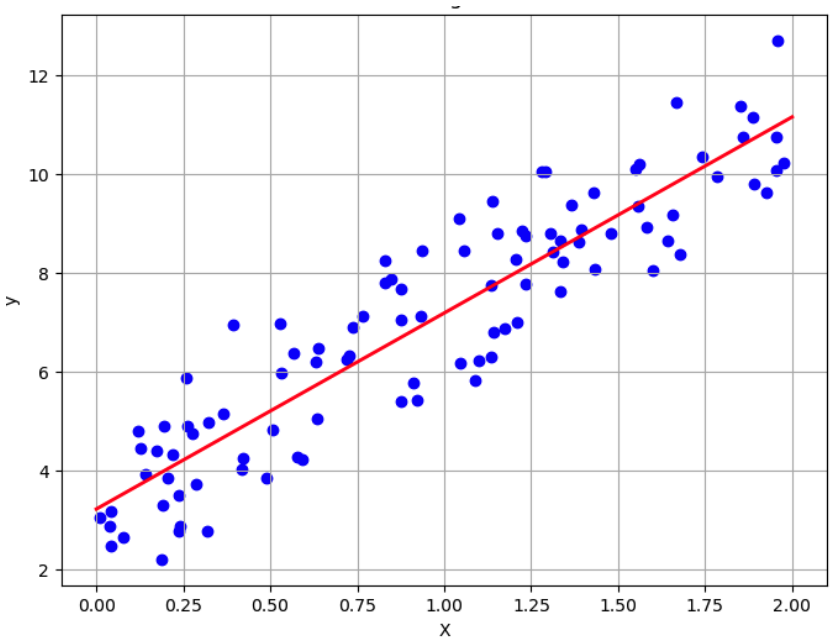
- Presented with a set of training instances
- Need to come up with a model (e.g., rule, pattern) that
  - distinguishes positive examples from negative ones, i.e., classifier
  - find a function for mapping data to targets, i.e., regression model



- **Training set**: a collection of instances from which a classifier is induced/trained
- **Test Set**: A collection of instances which were never used for learning the classifier
  - For measuring the performance of the learnt classifier

Machine Learning Tasks

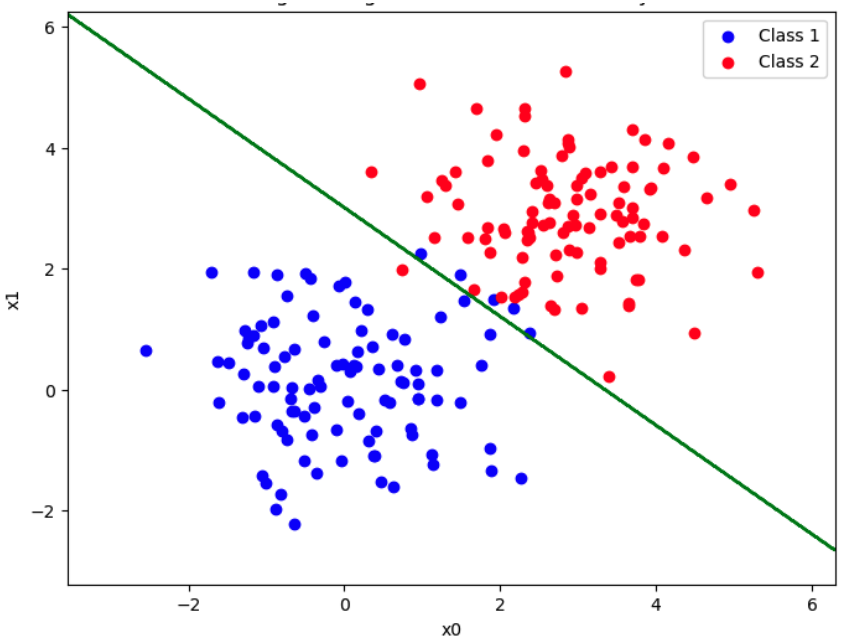
Regression



Map into real-valued prediction variable

- **Output:** a **continuous value**

Classification



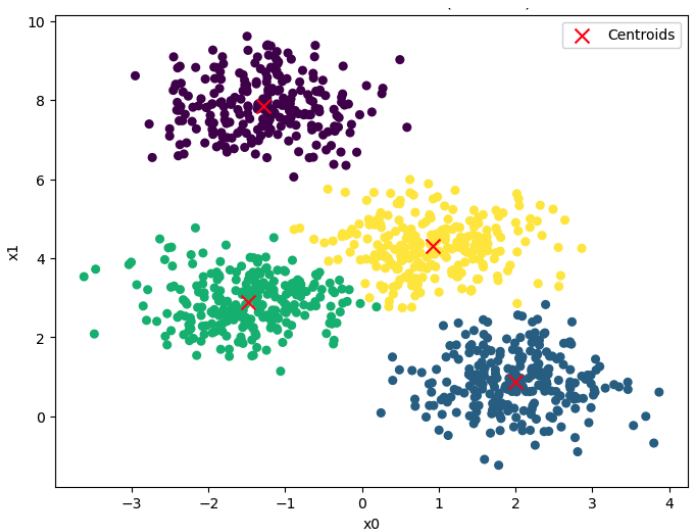
Maps into predefined groups/classes

- **Output**: possible class label

A regression task and classification task? depends on the “targets”

Clustering

- We still have **input data**, but there is **no ground-truth**
- There is no clear “right” answer, requires interpretation
- The goal is to discover some **hidden structures**, referred to as **clusters**



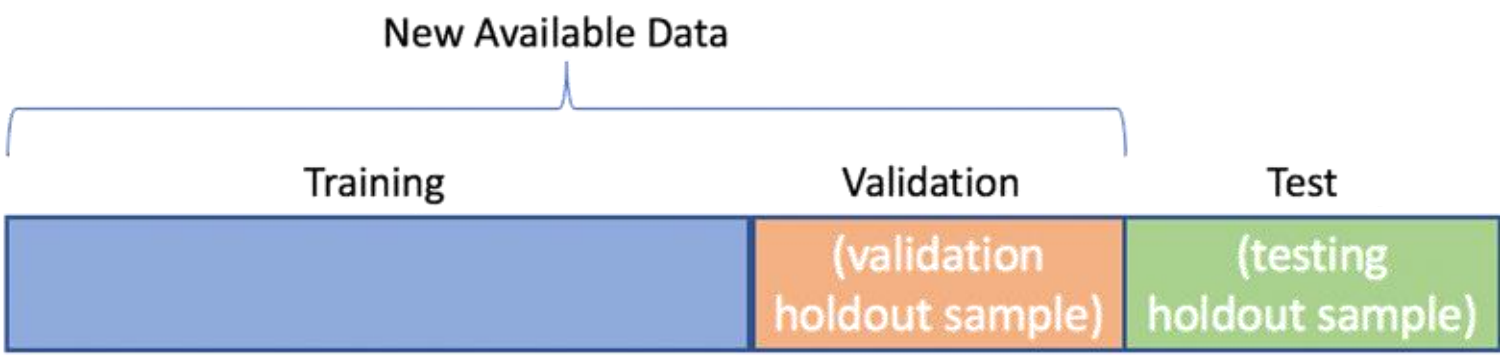
These are not classes!
Meaning is given to them according to an expert

Cluster != classes

Dataset Splitting

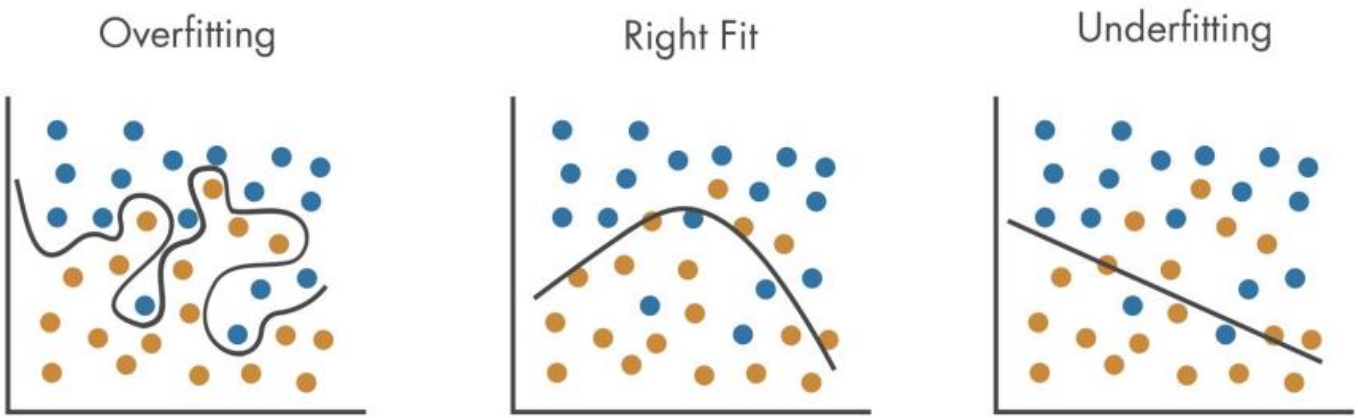
- Training set VS Test set

- Training set, **Validation set**, and Test set
  - Not used for training model and not used for test
  - Unseen data for learned model
  - But used during the training process, e.g., model selection, improve the process of training model



Generalisation

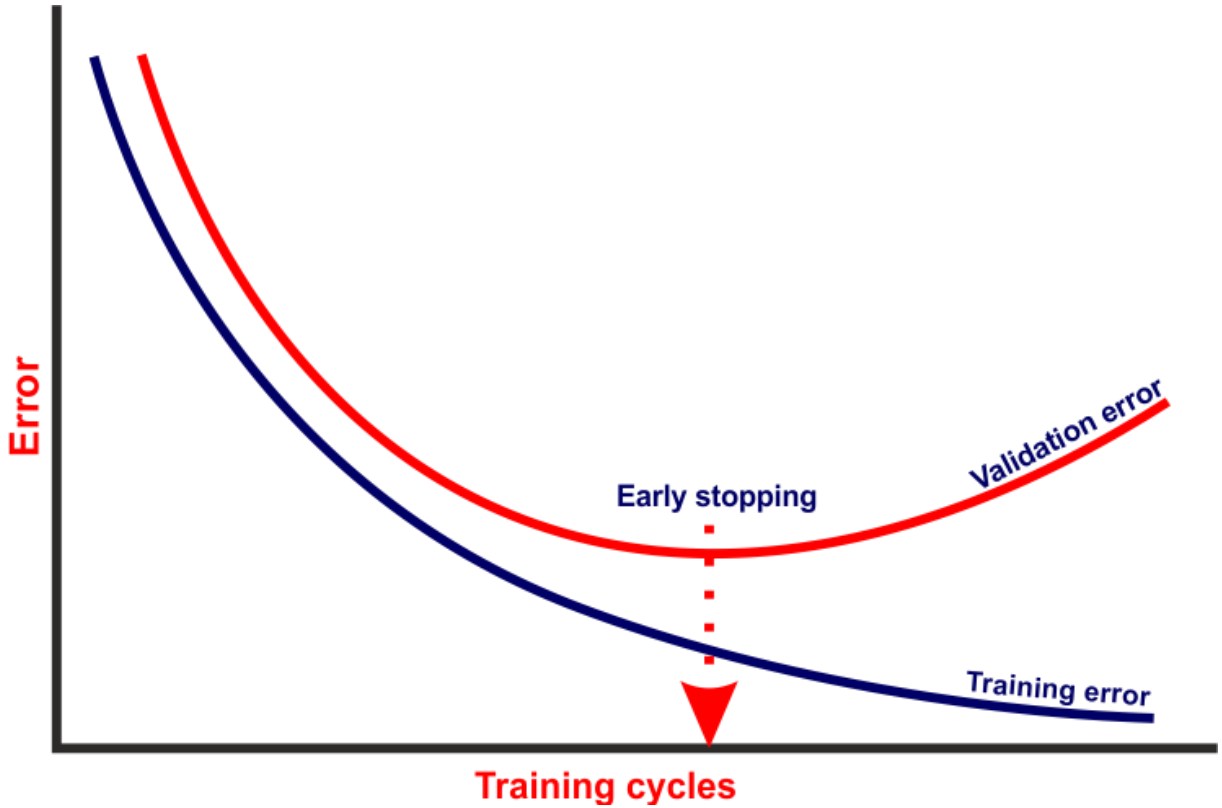
- We learn a classifier/predictor/model from the training data
- But performing well on training data is **NOT** enough!
- Important to evaluate the performance on the **test (unseen) data – generalisation**



Overfitting

How to handle overfitting:

- Early stopping
- Validation set



How Good the Model is?

- To measure the classification performance

Actual Values

Predicted	Positive (1)	Negative (0)
Positive (1)	TP	FP
Negative (0)	FN	TN

Confusion Matrix

TP: True Positive; FP: False Positive;
FN: False Negative; TN: True Negative

There are several Metrics: Accuracy, Precision, Recall, F1-score

$$\textit{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Overall classification accuracy

$$\textit{Recall} = \frac{TP}{TP + FN} \text{ or } \frac{\text{True Positive}}{\text{Actual Results}}$$

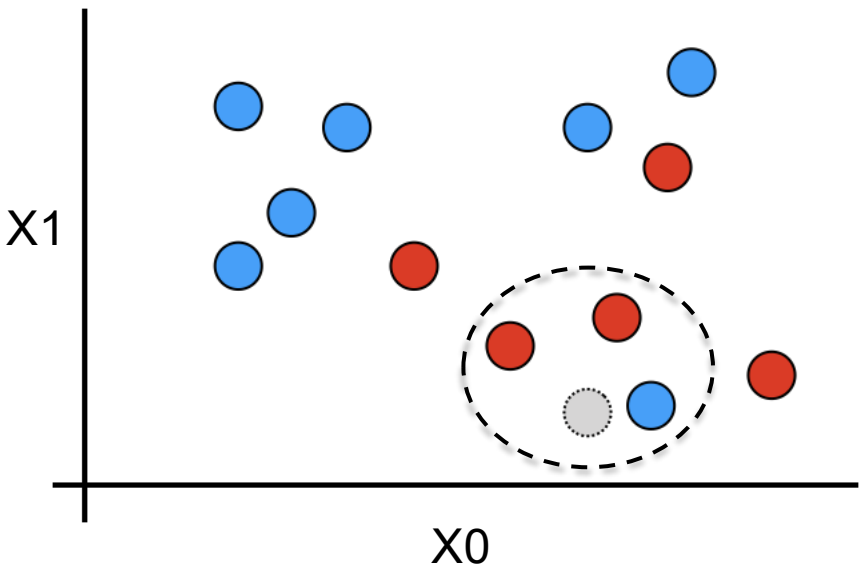
For the positive instances, how many of them are correctly predicted?

$$\textit{Precision} = \frac{TP}{TP + FP} \text{ or } \frac{\text{True Positive}}{\text{Predictive Results}}$$

Among predicted positive instances, how many of them are really positive?

$$\textit{F - score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

K-NN



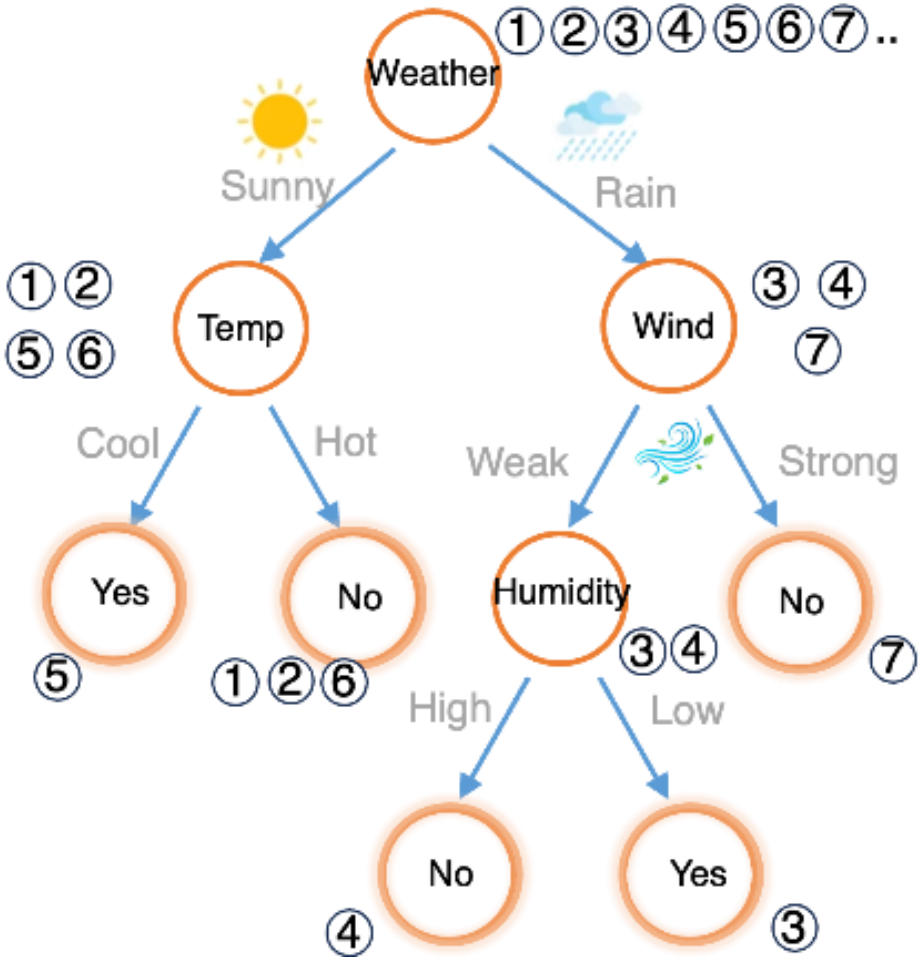
Assume a binary classification problem: blue and red circles

There are two input features X0 and X1

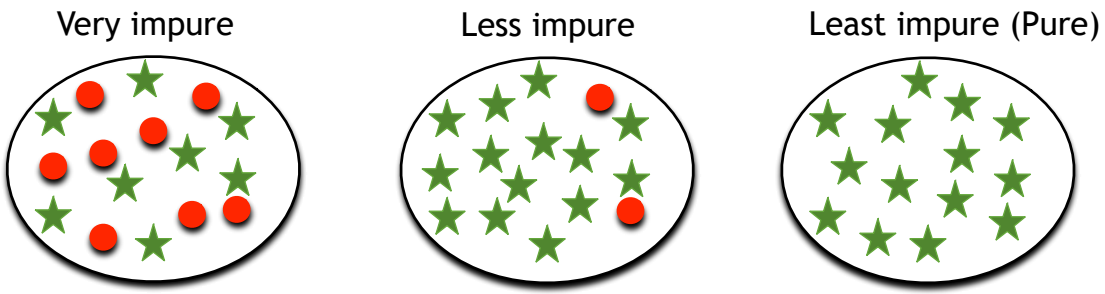
We want to classify the "unknown" gray instance

- **Training:** The training data is shown above (all the red and blue circles)
- We don't know the class for the gray instance.
- **Prediction:** Find the closest neighbor**s** in the training set to the gray inst.
  - Assuming **k = 3**, there are 1 blue and 2 red as nearest neighbors (prediction = red)

Build a Decision Tree



Pure or Impure?



- Impure->Non-Deterministic: bad
  - mixed classes in the node
  - need to ask more questions
- Pure->Deterministic: good
  - just one class in the node
  - can predict a class

Best Attribute: The splitting should make the child nodes as pure as possible

Node (im)purity: can be defined in different ways
- Probability-based
- Information theory based, i.e., Information Gain <--- Entropy

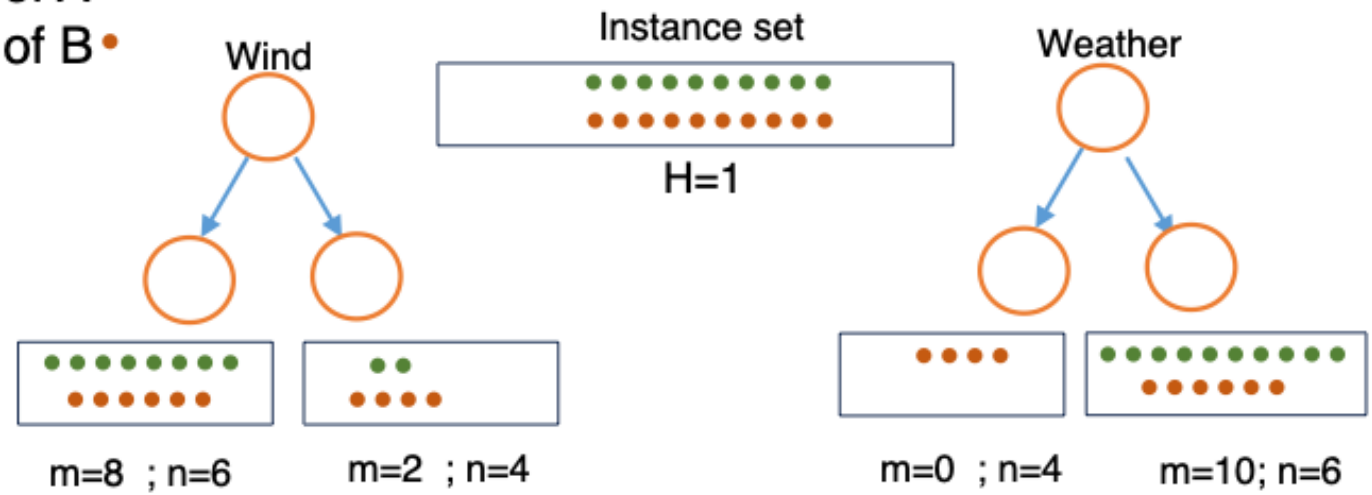
Choose a Good Attribute: Impurity of Nodes

$H(X) = -P(A)\log_2 P(A) - P(B)\log_2 P(B)$

Which attribute is better for splitting?

m: number of A •

n: number of B •



$$p(A) = \frac{8}{8+6} \approx 0.57$$

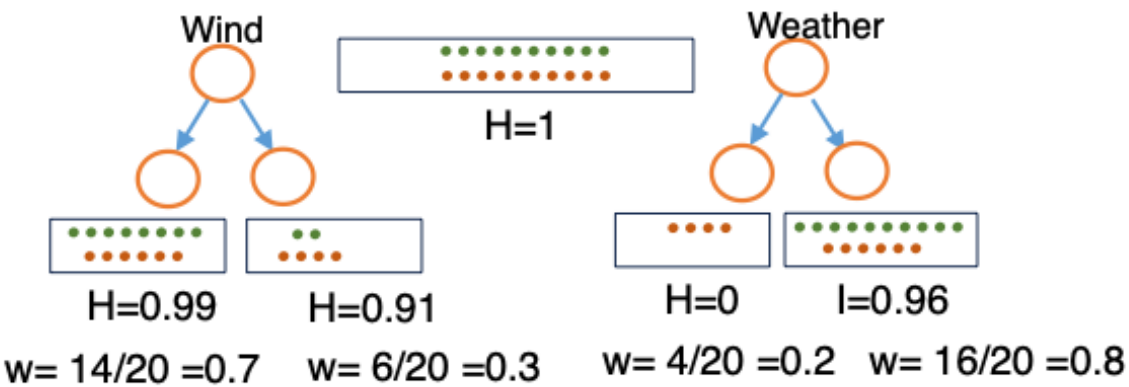
$$p(B) = \frac{6}{8+6} \approx 0.43$$

$$H(X) = -P(A)log_2P(A) - P(B)log_2P(B)$$

$$= -0.57 * log_2(0.57) - 0.43 * log_2(0.43)$$

=0.99

Choose a Good Attribute: Impurity of Nodes



- Goodness of an attribute: Compare the entropy before and after splitting:
- Information Gain (IG): $IG = H(parent) - \sum w_i H(child)$

$$IG(wind) = 1 - (0.7 * 0.99 + 0.3 * 0.91) = 1 - 0.966 = 0.034$$

$$IG(weather) = 1 - (0.2 * 0 + 0.8 * 0.96) = 1 - 0.768 = 0.232$$

$$IG(wind) < IG(weather)$$

Choose "Weather"

Three keys for Ensemble

Diversity, Combination, Base learner

Diversity: learners have to be “diverse” w.r.t their predictions

Combination: must suppress incorrect predictions and highlight correct predictions

Base learner: must be somewhat accurate

Inducing Diversity

Train on **different instances**: Bagging [1]

Train on **different features**: Random Subspaces [2]

Both: Random Forest [3] and Random Patches [4]

Heterogeneous learners: Combine different learners like decision trees, KNN, perceptrons, …

[1] Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

[2] Ho, Tin Kam. "The random subspace method for constructing decision forests." *IEEE TPAMI*, 1998

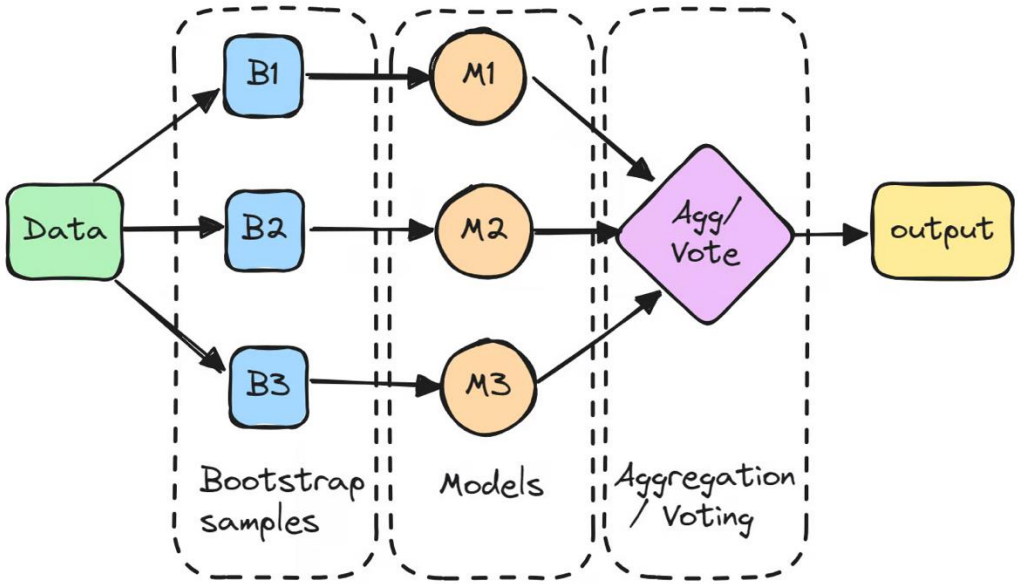
[3] Breiman, Leo. "Random forests." *Machine learning*, Springer, 2001

[4] Louppe, Gilles, and Pierre Geurts. "Ensembles on random patches." *ECML-PKDD 2012*

Bagging: Bootstrap Aggregation

Bagging trains each model of the ensemble with a bootstrap sample from the original dataset

- a **random** subset of the training data sample **with replacement**



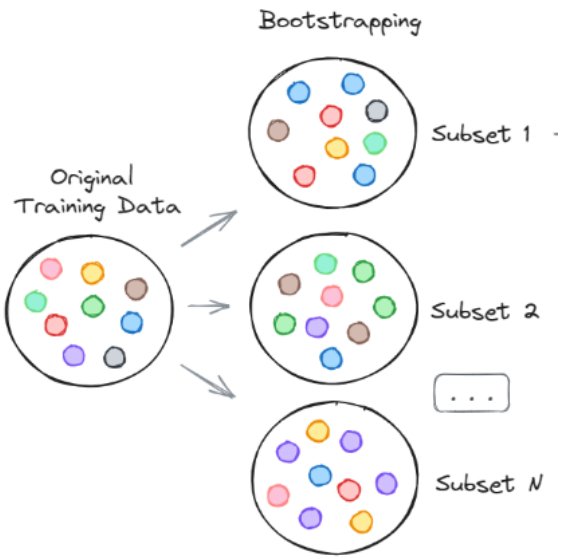
Breiman, Leo. "Bagging predictors." *Machine learning*, Springer, 1996.

Bagging– bootstrap sample

Sample **with replacement:** Sampling unit is returned before next is drawn

In the sample with replacement:

- Some are drawn into new dataset, some are left out
- Some are repeatedly drawn



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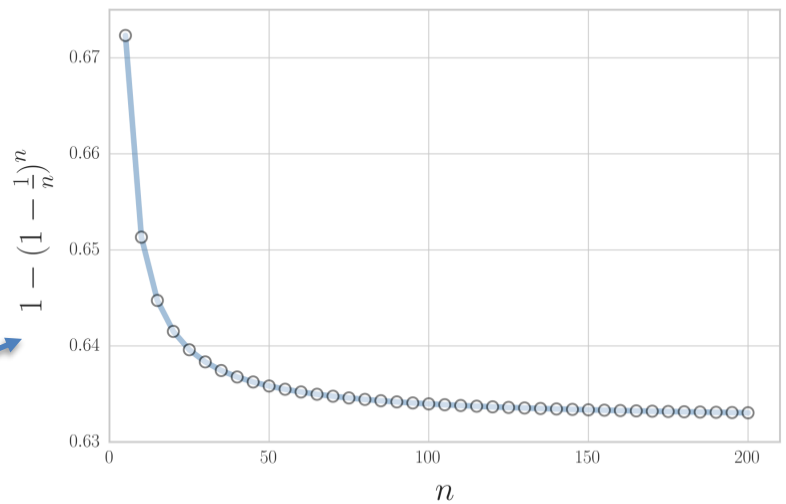
Bagging: subsamples

$$P(\textbf{not chosen}) = \left(1 - \frac{1}{n}\right)^n,$$

$$\frac{1}{e} \approx 0.368, \quad n \to \infty\,.$$

$$P(\textbf{chosen}) = 1 - \left(1 - \frac{1}{n}\right)^n \approx 0.632$$

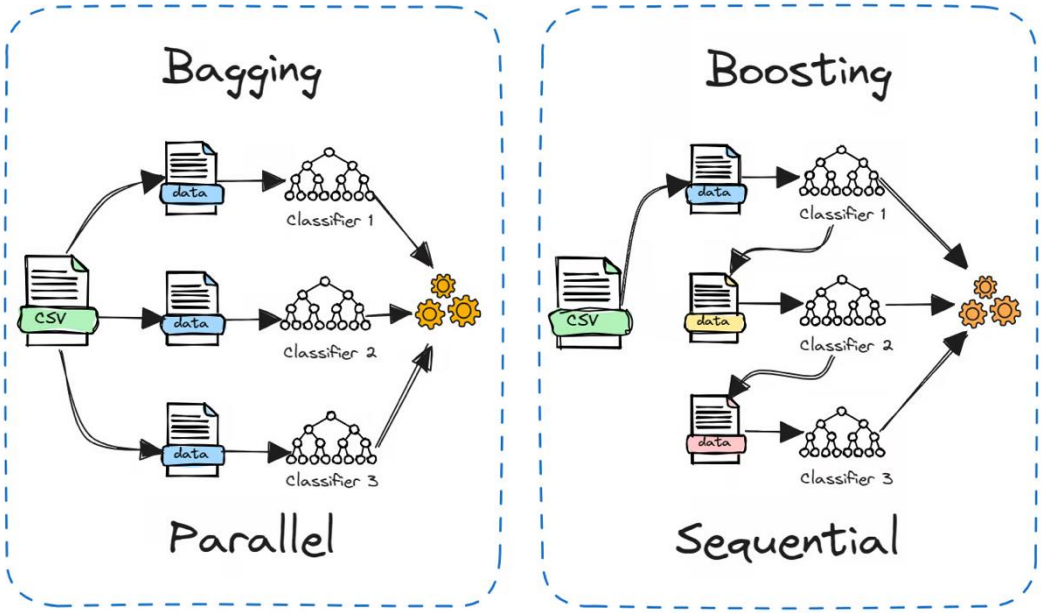


On average for each subsample:

- ~63.2% of the instances are from the original dataset
- ~36.8% are repeated instances
- ~36.8% of the original instances are not present

Boosting

- Boosting
  - Train classifiers sequentially, each time focusing on training examples/instances that the previous ones got wrong.
  - To focus on specific examples, boosting uses a weighted training set, i.e., a core mechanism where each data point is **assigned a specific weight** that is adjusted iteratively to force subsequent models to focus on the examples that previous models misclassified.



Boosting

- How to implement the “reweights”?

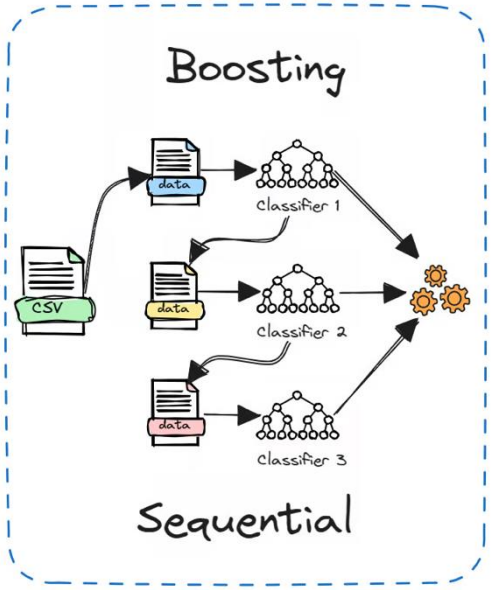
Method 1: consider into the total error

– If a data point has a large weight (because Classifier 1 missed it), misclassifying that single point might add 0.40 to the total error.

– Meanwhile, misclassifying a "light" point might only add 0.01 to the error.

Method 2: resampling the dataset

– Gives the instances that have higher weights larger probabilities to be selected



Boosting

- How to use the data?

  Method 1: use all data

  – Consider the weights into the total error

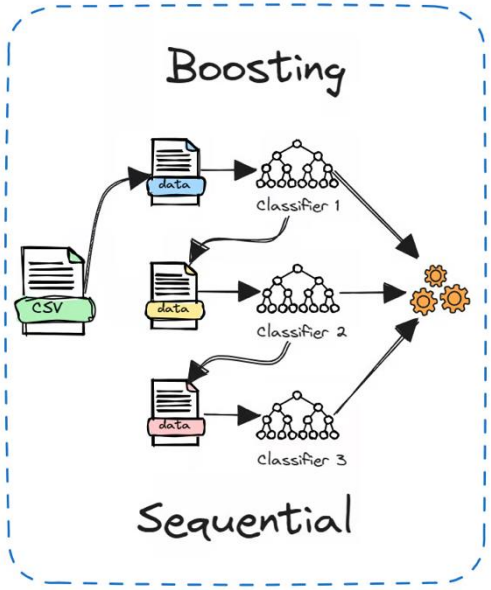
  Method 2: Resampling the Dataset

  – Gives the instances that have higher weights a larger probabilities to be selected,

    e.g., 1. Classifier 1 will be trained on a randomly selected 80% of your dataset;

    2. update weights, use *different* random 80% sample to train classifier 2

    3……



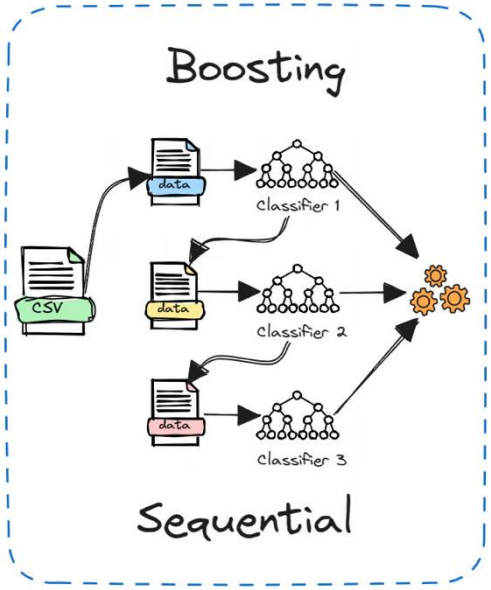
Boosting

- How to train classifiers?

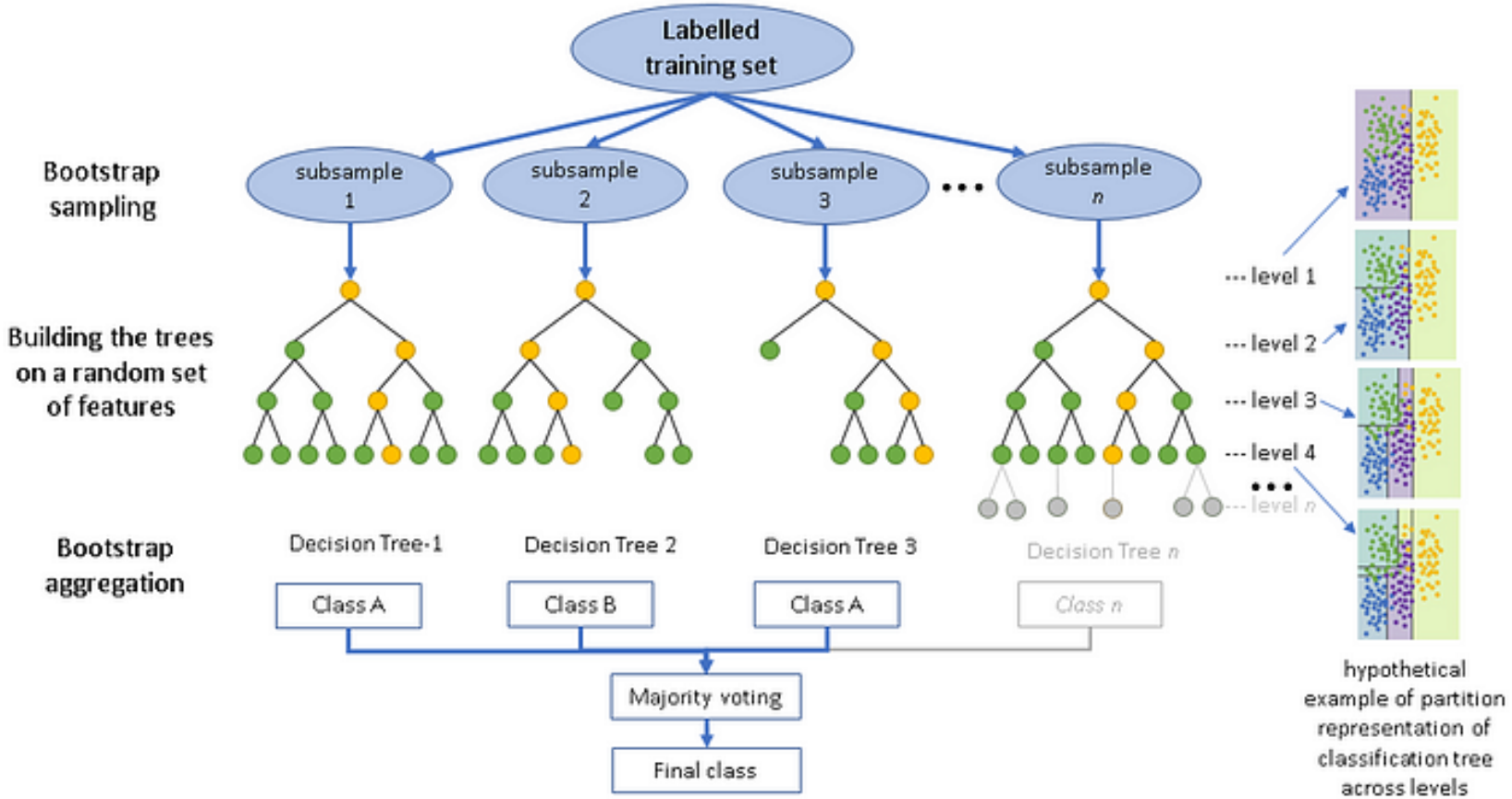
  Method 1: train each classifier separately (e.g., reweighted data)

  Method 2: incremental learning

  – E.g., classifier 2 is an improved version of classifier 1



Random Forest



Handling Missing Values

- Deletion approaches
  - Omits all records containing missing values. Only applies:
    - Missing data introduced in the MCAR mode,
    - When data contains less than 5% of missing values
- Imputation (estimation) approaches
  - Fill missing values with plausible values
    - Mean/Mode imputation
    - KNN imputation

Data Preparation

- **Farm data (meaning?)**

  0: "normal"

  1: "slightly pink"

  2: "a little pink"

  3: "a little bit pink"

  4: "pinkish"

  5: "pink"

chatGPT:

•**Normal** - This suggests no pink coloration.
•**Slightly pink** - Implies a very faint amount of pink, less than "a little pink."
•**A little pink** - Suggests a small amount of pink, but more than "slightly pink."
•**A little bit pink** - Very similar to "a little pink," but the wording suggests it might be just slightly less intense.
•**Pinkish** - Indicates something is pink-like but not fully pink, potentially more than "a little pink" but less than fully "pink."
•**Pink** - This would suggest a standard pink color, the most intense pink in this list.

Data Preparation

Features – leucocyte_appearance – HARD TO RANK
Normal
Occasional reactive lymphocyte present
Occasional reactive lymphocytes
Occasional reactive monocyte
Occasional reactive monocytes present
Reactive lymphocytes 1+ Reactive lymphocytes present 1+
Reactive monocytes present 1+
Reactive monocytes 3+ Reactive monocytes present 2+
Reactive monocytes present 2+, occasional reactive lymphocytes
Reactive monocytes present 3+ large numbers of lysed cells.
Lysis and degeneration of the leucocytes
Total wbc and differential are approximate due to lysis and degeneration of the leucocytes.

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Reasons people will say no

Privacy

Security

Exclusive commercial relations

Cost

Effort

Complexity

Benefit

Summary

- Supervised learning and unsupervised learning
- Regression, classification and clustering
- Dataset split: training, test and validation set
- Generalisation and overfitting
- KNN, decision tree, ensemble learning
- Data preparation